



Application of a new empirical model of nonlinear self-excited force to torsional vortex-induced vibration and nonlinear flutter of bluff bridge sections

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ABSTRACT

Long-span bridges are susceptible to vortex-induced vibration (VIV) and flutter in different reduced wind range. Currently, VIV and flutter are considered as two different types of wind-induced vibration and different empirical models were proposed. The possibility of building a universal empirical model for both torsional VIV and nonlinear flutter was explored. In the previous study, a new empirical model was proposed for describing aerodynamic nonlinearities during soft flutter of a bluff bridge section with a medium side ratio. In this study, the empirical model was refined and extended to both torsional VIV and nonlinear flutter for a general bluff body, especially for a flat bridge deck, which is more common in long-span bridges. In the empirical model, aerodynamic nonlinear damping effect was modeled by a cubic velocity term and its applicability was validated by a series of elastically-mounted sectional model tests of a flat twin-side-girder bridge deck. The results indicated that the proposed empirical model was suitable for predicting the stable amplitude of torsional VIV and nonlinear flutter, although the motion-induced “pure force” varies with side ratios. It was found that torsional VIV and soft flutter share the same nonlinear damping mechanism during the development of LCO.

1. Introduction

Long-span cable-supported bridges with bluff section are prone to wind actions and suffer from the risk of aeroelastic instabilities, which arise from the interaction between the bridge section and its surrounding air. During past decades, the aeroelastic instabilities of flexible bridges have been extensively investigated and are conventionally divided into two basic types, i.e., vortex-induced vibration (VIV) and flutter.

VIV is induced by the periodic vortex-shedding from the outer surface of bridge deck and the resultant fluctuating force was characterized by a dominant dimensionless frequency called as Strouhal number. Although vertical VIV phenomena are more common in-situ observations, torsional VIVs have been reported in the preliminary design of many bridges in wind tunnel tests and also slender members with bluff section, such as H-shape hanger of Dongping Bridge by Chen et al. (2012), a twin-side-girder bridge deck of Yingwuzhou Bridge. VIV is a nonlinear aeroelastic phenomenon and characterized by a limit-cycle oscillation

(LCO) in the time domain, the variation of vibration amplitude with wind speed manifest as a “lock-in” phenomenon. To accurately predict vortex-induced vibration responses, the characteristics of fluctuating vortex-induced force of typical bridge sections have been investigated and many mathematical models of vortex-induced force were proposed, which consists of single-degree-of-freedom (SDOF) models (Ehsan and Scanlan, 1990; Larsen, 1995; Zhu et al., 2017) and 2-DOF models (Hartlen and Currie, 1970; Diana et al., 2006). Recently, Zhu et al. (2017) directly measured the time history of nonlinear vortex-induced force by a novel force measurement technique and proposed a Rayleigh-type model of vertical vortex-induced force. The Van der Pol-type SDOF model proposed by Ehsan and Scanlan (1990) was not suitable for vertical VIV of a typical closed-box bridge deck. The pure vortex shedding force in the total vortex-induced force was also found to play a negligible role on VIV responses and VIV can thus be treated as a nonlinear self-excited vibration (Ehsan and Scanlan, 1990; Zhu et al., 2017).

Flutter occurs at higher wind velocity than VIV and is characterized by torsional vibration coupled with heaving DOF. The heave-torsion

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Nomenclature

a_α	instantaneous torsional amplitude (rad)
A_0	initial amplitude (–)
A_2^*	aeroelastic parameter (–)
A_3^*	aeroelastic parameter (–)
$\tilde{A}_2^*$	fitted aeroelastic parameter (–)
$\tilde{A}_3^*$	fitted aeroelastic parameters (–)
B	width of a bridge deck (m)
$\tilde{c}_m$	structural damping coefficient (N's)
$\tilde{C}_L$	dimensionless lift coefficient (–)
$\tilde{C}_M$	dimensionless torque coefficient (–)
D	depth of a bridge deck (m)
E	dimensionless mechanical energy (–)
f	vibration frequency (Hz)
f_t, f_{t0}	torsional frequency in still air (Hz)
h	heaving displacement (m)
$\dot{h}$	heaving velocity (m/s)
I	mass moment of inertia per unit length (kg m ² /s)
I_r	mass ratio (–)
J_m	mass moment of inertia (kg m ²)
$\tilde{k}_m$	structural stiffness coefficient (N)
K	reduced frequency (–)
K_s	reduced natural frequency (–)
K_{vs}	non-dimensional vortex shedding frequency (–)
L_{se}	aerodynamic lift per unit length (N/m)

M_{se}	aerodynamic torque per unit length (N'm/m)
$\tilde{M}_{se}$	aerodynamic torque per unit length (–)
$\widehat{M}_{se}$	measured aerodynamic torque per unit length (–)
s	dimensionless time (–)
t	time (s)
U	oncoming flow velocity (m/s)
U^*	reduced velocity (–)
W_{ξ_s}	accumulative work done (–)
Y_1^*	aeroelastic parameter (–)
Y_3^*	aeroelastic parameter (–)
α	torsional displacement (rad)
$\dot{\alpha}$	torsional velocity (rad/s)
$\hat{\alpha}'$	dimensionless angular velocity (–)
α'	dimensionless angular velocity (–)
α''	dimensionless angular acceleration (–)
α_N	parameter of asymptotic amplitude (–)
β	stable dimensionless amplitude (–)
ε	aeroelastic parameter (–)
ε_{ij}	aeroelastic parameter (–)
$\tilde{\varepsilon}_{03}$	fitted aeroelastic parameters (–)
ξ_s	structural damping ratio (–)
ξ_{s0}	structural damping ratio in still air (–)
φ	non-dimensional phase difference (–)
ψ_0	initial phase (–)
ω_{t0}	torsional frequency in still air (rad/s)
ω_s	undamped torsional frequency (Hz)

coupling effect varies with different aerodynamic configurations (Liao et al., 2019). For bluff sections with a large frontal area such as a two-edge-girder section, flutter usually exhibits less coupling than for streamlined bridge decks such as a closed-box section. Flutter is conventionally described by a classical linear flutter theory developed from Scanlan's linear self-excited force model (Scanlan and Tomko, 1971). On the basis of small-vibration amplitude, classical linear flutter theory simplifies flutter as a complex eigenvalue problem and predicts flutter response as a sudden increase of vibration amplitude beyond flutter boundary, therefore, traditional linear flutter is often called as "hard flutter" for its amplitude will diverge infinitely with time when wind speed is larger than flutter onset point. In recent years, the nonlinear aspect of flutter instability has attracted extensive attentions (Diana et al., 2010; Náprstek et al., 2007, 2010; Ying et al., 2017; Gao et al., 2018, 2020a; Liao et al., 2019). Experimental evidences suggest that bridge decks with bluff configurations or under relatively large attack angle will reach a secondary re-stabilization due to aerodynamic nonlinearities and undergo nonlinear "soft flutter" phenomenon for its amplitude will not diverge infinitely with time beyond flutter boundary but will develop to a stable one, where "soft" means that the curve of the stable amplitude vs. wind speed is smooth and without any evident break-point (Liao et al., 2019). Several empirical models have been developed to mathematically model the nonlinear self-excited force during soft flutter, such as the nonlinear Rheologic model by Diana et al. (2010), a nonlinear convolution scheme by Wu and Kareem (2013), the nonlinear 2-DOF coupled model by Náprstek et al. (2007, 2011) and Gao et al. (2020b). Recently, the authors investigated the nonlinear characteristics of soft flutter of a twin-side-girder section with a medium side ratio as $B/D = 7.8$ and proposed an nonlinear empirical model to estimate the stable amplitude of soft flutter by incorporating a nonlinear Rayleigh-type damping term into the classical linear model (Gao et al., 2018).

As mentioned above, VIV and flutter are conventionally considered as two different types of wind induced oscillation for flexible long-span bridges. However, experimental evidences demonstrated that VIV and

soft flutter share in fact many common features. VIV and soft flutter arise from the same physical mechanism which is the nonlinear interaction of bluff section with flowing air; both of them can thus be treated as a nonlinear self-excited vibration and both exhibit a limit cycle oscillation (LCO) in the time history of vibration responses. The differences between VIV and soft flutter are that VIV exhibits "lock in" phenomenon of amplitude with wind speed, whereas soft flutter demonstrates a gradual increase of amplitude vs. wind speed with no obvious upper limit (Gao et al., 2018, 2020a). The above different phenomenon of VIV and soft flutter can be viewed as the aeroelastic characteristics in different ranges of reduced velocity. Therefore, there is a possibility to build a universal mathematical model of nonlinear self-excited force for both torsional VIV and soft flutter.

In this study, the possibility of building a universal model of nonlinear self-excited force was explored preliminarily in the domain of SDOF torsional vibration for bluff bridge sections with relatively large frontal area, for which flutter exhibits less heave-torsion coupling and can be simplified as SDOF torsional vibration. Firstly, the previously proposed empirical model based on the soft flutter of a medium-side-ratio section was refined and extended to an empirical model of both torsional VIV and soft flutter. The universal asymptotic solution of nonlinear self-excited vibration and identification of nonlinear aerodynamic parameters were then proposed. Finally, the applicability of the empirical model was estimated by a series of elastically-mounted sectional model tests on a flat twin-side-girder section with a side ratio larger than 10 by using a novel measurement technique of nonlinear aerodynamic force.

2. SDOF nonlinear self-excited force model

2.1. Nonlinear empirical model

The governing equation of torsional aeroelastic vibration for an elastically-supported bridge deck immersed in flowing air can be expressed as

$$I(\ddot{\alpha} + 2\xi_s \omega_s \dot{\alpha} + \omega_s^2 \alpha) = M_{se}(\alpha, \dot{\alpha}, h, \dot{h}, U, t) \quad (1)$$

where α and $\dot{\alpha}$ are the torsional displacement and velocity, respectively. t represents time. I is the effective mass moment of inertia per unit length. ω_s is the undamped vibration frequency and ξ_s is the structural damping ratio. h and $\dot{h}$ are the heaving displacement and velocity; M_{se} is aerodynamic torque per unit length due to the interaction between flowing air and the deck motion, which is dependent on oncoming flow velocity U and vibration responses, i.e., $\alpha, \dot{\alpha}, h, \dot{h}$.

For a bluff bridge section, such as a twin-side-box section or a π -shape section, the observed torsional VIV and soft flutter were found to occur in the torsional mode with slight heave-torsion coupling effect (Gao et al., 2018; Zhang et al., 2019). The dependence of M_{se} on the coupling of heaving DOF in Eq. (1) can thus be ignored, that is

$$I(\ddot{\alpha} + 2\xi_s \omega_s \dot{\alpha} + \omega_s^2 \alpha) = M_{se}(\alpha, \dot{\alpha}, U, t) \quad (2)$$

Eq. (2) can be further expressed in a dimensionless form as

$$\alpha'' + 2\xi_s K_s \alpha' + K_s^2 \alpha = \tilde{M}_{se}(\alpha, \alpha', s) \quad (3)$$

Where α' and α'' represent the dimensionless angular velocity and acceleration, respectively, $K_s = \omega_s B/U$ is the reduced natural frequency and B is the width of bridge deck; $s = tU/B$ is dimensionless time and $\tilde{M}_{se} = M_{se} B^2/(IU^2)$.

Challenge lies in the empirical modelling of $M_{se}(\alpha, \dot{\alpha}, U, t)$. Investigators have proposed empirical models in the form of SDOF (Ehsan and Scanlan, 1990; Larsen, 1995; Zhu et al., 2017) or 2-DOF (Hartlen and Currie, 1970; Diana et al., 2006) for VIV and a few nonlinear models of soft flutter (Náprstek et al., 2007, 2011; Diana et al., 2010; Gao et al., 2018, 2020b). Among the existing models, the SDOF model proposed by Ehsan and Scanlan (1990) originally for vertical VIV was widely used. Scanlan's model was further extended to torsional VIV by some investigators (Xian et al., 2009) which is expressed as

$$\tilde{M}_{se} = I_r \left[KA_2^* (1 + \varepsilon \alpha^2) \alpha' + K^2 A_3^* \alpha + \frac{1}{2} \tilde{C}_M \sin(K_{vs} s + \varphi) \right] \quad (4)$$

where $I_r = \rho B^4/I$ is the mass ratio, $\tilde{C}_M$ is the dimensionless torque coefficient of pure vortex shedding, $K = \omega B/U$ is reduced frequency. A_2^*, ε and A_3^* are K -dependent aeroelastic parameters to be identified through wind tunnel tests. $K_{vs} = \omega_{vs} B/U$ and φ are the non-dimensional frequency and phase difference of vortex shedding. The term $\tilde{C}_M \sin(K_{vs} s + \varphi)$ is related with the instantaneous torque by pure vortex shedding from a bluff body.

As can be seen in Eq. (4), Scanlan's model describes nonlinear self-excited force during VIV by a nonlinear aerodynamic damping term, which has a nonlinear characteristic of Van der Pol oscillator. Also note that Scanlan's model was recently extended to soft flutter by Chen and Ma (2011) by dropping $\tilde{C}_M \sin(K_{vs} s + \varphi)$ from Eq. (4).

Using a newly developed measurement technique, the authors found from previous study (Gao et al., 2018) that Scanlan's model as Eq. (4) cannot satisfactorily fit the measured $M_{se}(t)$ during soft flutter of a bluff bridge deck. A new empirical model of nonlinear self-excited torque was proposed to estimate the stable amplitude of soft flutter as

$$\tilde{M}_{se} = I_r [KA_2^* (1 + \varepsilon_{03} \alpha'^2) \alpha' + K^2 A_3^* (1 + \varepsilon_{30} \alpha^2) \alpha] \quad (5)$$

where ε_{03} and ε_{30} are K -dependent aerodynamic parameters.

In the new model as Eq. (5), the nonlinear self-excited torque is modeled by a linear damping term α' , a cubic nonlinear damping term α'^3 , a linear stiffness term α and a nonlinear stiffness term α^3 . The difference between the proposed model and Scanlan's model lies in the choice of nonlinear damping terms.

To extend to the case of torsional VIV, the empirical model as Eq. (5) is refined as

$$\tilde{M}_{se} = I_r \left[KA_2^* (1 + \varepsilon_{03} \alpha'^2) \alpha' + K^2 A_3^* \alpha + \frac{1}{2} \tilde{C}_M \sin(K_{vs} s + \varphi) \right] \quad (6)$$

The nonlinear empirical model as Eq. (6) incorporates the term $\tilde{C}_M \sin(K_{vs} s + \varphi)$ to consider the effect of pure vortex-shedding force for the case of small amplitude vibration during VIV. According to Vickery and Basu (1981, 1983), the pure vortex shedding term plays an important role in the estimation of random vibrations under small amplitudes and also necessary to account for the influence of turbulent flow. The VIV responses of bluff cylinders can be divided into three parts depending on the relative values of non-dimensional mass-damping parameters K_s and K_{a0} , which are defined as

$$K_s = \frac{m \xi_s}{\rho D^2}, \quad K_{a0} = \frac{m \xi_{a0}}{\rho D^2} \quad (7)$$

where m is the mass per unit length, ξ_s is the structural damping ratio, ξ_{a0} is the aerodynamic damping ratio under small amplitude, ρ is air density and D is the characteristic length. The first part is the "lock-in" region corresponding approximately to $K_s/K_{a0} < 0.8$. In this region, nonlinear aeroelastic damping dominates the vibration responses and the pure vortex shedding term plays a negligible role. The vibration response is sinusoidal. The third part is a "forced vibration" region corresponding approximately to $K_s/K_{a0} > 1.2$. In this range, the effect of aeroelastic nonlinearity is ignorable and the motion can be treated as a forced random vibration which is Gaussian. The second part is the "transition" region. In this region, both the random pure vortex shedding and the nonlinear aerodynamic damping play significant roles. The vibration responses changes from Gaussian to sinusoidal with decreasing K_s . In addition, the pure vortex shedding force term is necessary when predicting the random vibrations due to turbulent flow as suggested by Vickery (1981). Vickery and Basu (1983) modeled the large-scale turbulent effect on the across-flow vibration of slender structures by the spectrum of random lift coefficient and relating the bandwidth of fluctuating lift force with turbulence intensity.

It should be pointed out that the applicability of the nonlinear motion-induced force in Eq. (6) was based on the observed soft flutter phenomenon of a twin-side-girder section with a medium side-ratio as $B/D = 7.8$ in the previous study (Gao et al., 2018), however, it is still unknown whether Eq. (6) is suitable for torsional VIV or soft flutter for bluff sections with other side ratios, especially for larger side ratios, because a flat section is more common in long-span bridges. On the other hand, Zhu et al. (2015) suggested that for a centrally-slotted-box section the nonlinear self-excited torque during torsional VIV was described by Scanlan's model instead of Eq. (6). Whereas, as discussed above, torsional VIV and soft flutter are both nonlinear aeroelastic instabilities and it is reasonable to assume that they share the same form of aerodynamic nonlinearities for similar aerodynamic configurations regardless of the side-ratio variation. Therefore, the proposed model is extended to be a universal empirical model of nonlinear self-excited torque of torsional VIV and soft flutter for bluff bridge sections with a large frontal area such as a twin-side-girder section, a π -shape section, etc., for which wind tunnel tests are needed to further check its applicability.

2.2. Asymptotic solution of self-excited vibration

Based the proposed empirical model, the non-dimensional governing equation of torsional VIV and soft flutter is obtained by substituting Eq. (6) into Eq. (3):

$$\alpha'' + 2\xi_s K_s \alpha' + K_s^2 \alpha = I_r \left[KA_2^* (1 + \varepsilon_{03} \alpha'^2) \alpha' + K^2 A_3^* \alpha + \frac{1}{2} \tilde{C}_M \sin(K_{vs} s + \varphi) \right] \quad (8)$$

Similar to Ehsan and Scanlan (1990), an approximate analytical solution can be obtained by neglecting the pure vortex shedding force term. The asymptotic solution of Eq. (8) is deduced under the assumption that

the aeroelastic nonlinearity is weak compared with the elastic restoring force in the spring-suspended oscillatory system, which results in a quasi-harmonic torsional response with slow-varying amplitude and phase (Gao et al., 2018). Therefore, the method of averaging or multi-scales (Nayfeh and Mook, 1979; Gao and Zhu, 2015) can be employed to obtain a first-order asymptotic solution of Eq. (8) as follows

$$\alpha(s) = \frac{\beta}{\sqrt{1 - \left(\frac{A_0^2 - \beta^2}{A_0^2}\right) e^{-(\alpha_N \beta^2/4)s}}} \cos \left\{ Ks - \frac{1}{2K} [I_r K^2 A_3^* + (K^2 - K_s^2)] s + \psi_0 \right\} \quad (9)$$

$$\alpha_N = -3I_r A_2^* K^3 \varepsilon_{03} \quad (10)$$

$$\beta = \frac{2}{K \sqrt{-3\varepsilon_{03}}} \left(1 - \frac{2\xi_s K_s}{I_r A_2^* K} \right)^{1/2} \quad (11)$$

where A_0 and ψ_0 are the initial amplitude and phase.

The stable amplitude of LCOs during aeroelastic instabilities can be yielded as

$$\alpha(+\infty) = \beta = \frac{2}{K} \sqrt{\frac{2\xi_s K_s - I_r A_2^* K}{3I_r A_2^* K \varepsilon_{03}}} \quad (12)$$

It is indicated from Eq. 9–12 that the incorporation of nonlinear aerodynamic damping in the empirical model as Eq. (6) leads to a slow modulation of vibration amplitude. The stable amplitude of soft flutter and VIV are dependent on the relative value of structural damping, linear aerodynamic damping and nonlinear aerodynamic damping.

2.3. Identification of aerodynamic parameters

Since the flow field around a bluff body is very complicated during VIV and soft flutter, it is difficult to determine the theoretical values of the aerodynamic parameters in Eq. (6). The aerodynamic parameters in an empirical model are conventionally estimated from the vibration responses of elastically-mounted sectional model tests, the identified parameters are then applied to prototype bridges.

The pure vortex shedding coefficients $\bar{C}_M$ in Eq. (6) can be measured from a stationary sectional model. As for the aerodynamic parameters related to motion-induced force, Ehsan and Scanlan (1990) developed an identification method of aerodynamic force via “Grow to Resonance (GTR)” and “Decay to Resonance (DTR)” responses of a sectional model in free vibration tests based on the asymptotic solution. Zhu et al. (2017) used a direct nonlinear-least-square-fitting of the measured vortex-induced force to identify the aerodynamic parameters in their proposed model. To enhance the identification precision, an “energy-balanced principle” was developed based on the instantaneous balance of mechanical energy flow in the elastic oscillatory system.

The proposed identification method is based on the fact that aerodynamic damping force is very small compared with the aerodynamic stiffness force and other “pure force” components, as a result, the aerodynamic damping parameters are not sensitive to the residual of a least square fitting of aerodynamic force. Whereas, the non-conservative forces, i.e., aerodynamic damping and structural damping, are responsible for the mechanical energy dissipation; aerodynamic damping parameters are hence sensitive to the variation of mechanical energy and can be estimated by fitting the instantaneous work done by the aerodynamic force, that is

$$\int_{t_1}^t \widehat{M}_{se}(\tau) \cdot \widehat{\alpha}'(\tau) d\tau = K A_2^* I_r \left[\int_{t_1}^t \widehat{\alpha}'(\tau)^2 d\tau + \varepsilon_{03} \int_{t_1}^t \widehat{\alpha}'(\tau)^4 d\tau \right] \quad (13)$$

where the variables with $\widehat{}$ denote the measured values from a sectional model test.

Note that aerodynamic stiffness term and pure vortex shedding force term in Eq. (6) do not appear on the right hand side of Eq. (13), because their accumulative work are both very small during the development of amplitude and thus can be neglected.

On the other hand, the conservative forces, i.e., aerodynamic stiffness, inertia force and elastic restoring force, contribute only to the transference of mechanical energy between kinetic energy and elastic potential energy in the elastic oscillatory system, therefore, aerodynamic stiffness parameters are sensitive to the amount of transferred mechanical energy and thus can be identified by fitting the reactive work done by the self-excited torque, that is

$$\int_{t_1}^t \widehat{M}_{se}(\tau) \widehat{\alpha}(\tau) d\tau = K^2 A_3^* I_r \int_{t_1}^t \widehat{\alpha}(\tau)^2 d\tau \quad (14)$$

The aerodynamic damping and stiffness parameters in the proposed model can then be identified via a two-step nonlinear-least-square-fitting based on Eq. (13) and Eq. (14), respectively. Note that Eq. (13) and Eq. (14) need measured signals of self-excited force, for the cases where only the time history of vibration displacement is available, the above identification method can be used by further considering the following conditions.

Integrate the both sides of governing equation (3) with respect to torsional angle, yields

$$\begin{aligned} \int_{t_1}^t M_{se}(\tau) \alpha'(\tau) d\tau &= \int_{t_1}^t (\alpha'' + 2\xi_s K_s \alpha' + K_s^2 \alpha) d\alpha = \frac{1}{2} (\alpha'^2 + K_s^2 \alpha^2) \Big|_{t_1}^t \\ &+ \int_{t_1}^t 2\xi_s K_s \alpha'^2 d\tau = E(t) - E(t_1) + W_{\xi_s} \end{aligned} \quad (15)$$

where $E(t) = 0.5(\alpha'^2 + K_s^2 \alpha^2)$ represents the dimensionless mechanical energy of the oscillatory system, and $W_{\xi_s} = \int_{t_1}^t 2\xi_s K_s \alpha'^2 d\tau$ is the accumulative work done by structural damping force. Dimensionless angular velocity α' can be calculated by a finite difference operation from the measured torsional displacement.

Similarly, the integration $\int_{t_1}^t \widehat{M}_{se}(\tau) \cdot \widehat{\alpha}(\tau) d\tau$ in Eq. (14) can be obtained from governing Eq. (3) as

$$\int_{t_1}^t M_{se}(\tau) \alpha(\tau) d\tau = \int_{t_1}^t (\alpha'' + 2\xi_s K_s \alpha' + K_s^2 \alpha) \alpha d\tau \quad (16)$$

According to Eq. (9), torsional angle is simplified by neglecting slow-modulations of phase angle as $\alpha(\tau) = A(s) \cos(Ks + \psi_0)$. Then substitute $\alpha(\tau) = A(s) \cos(Ks + \psi_0)$ into Eq. (16) and one yields

$$\int_{t_1}^t M_{se}(\tau) \alpha(\tau) d\tau = \frac{1}{2} \int_{t_1}^t A(\tau) [A''(\tau) - K^2 A(\tau) + K_s^2 A(\tau) + 2\xi_s K_s A'(\tau)] d\tau \quad (17)$$

Eq. (15) and Eq. (17) reflect the principle of energy conservation during aeroelastic vibration and can be substituted into Eq. (13) and Eq. (14) to replace $\int_{t_1}^t \widehat{M}_{se}(\tau) \cdot \widehat{\alpha}'(\tau) d\tau$ and $\int_{t_1}^t \widehat{M}_{se}(\tau) \cdot \widehat{\alpha}(\tau) d\tau$, respectively, when the signals of aerodynamic force are not available.

3. Validation with wind tunnel tests

3.1. Sectional model tests

A flat bluff bridge deck with a side-ratio as large as 11.9 was adopted to check the applicability of the proposed empirical model to torsional VIV and nonlinear flutter in the case of a large side ratio. The wind tunnel tests of torsional VIV and soft flutter were conducted in TJ-2 Wind Tunnel with a testing section of 2.5m in width and 3m in height. As shown in Fig. 1, the sectional model was suspended between two fairing walls, which were made of arc shape at windward side ends to improve the flow quality. The two fairing walls were installed adjacent to the sectional



Fig. 1. Sectional model suspended between two fairing walls in TJ-2 Wind Tunnel.

model and acted as two large end plates to suppress the three dimensional vortex shedding around the ends of the sectional model.

The sectional model was manufactured by simulating the deck of Jingsha Bridge at a length scale of 1/25. Jingsha Bridge is a pre-stressed concrete (PC) cable-stayed bridge with a main span of 500m located in Hubei province of China. Fig. 2 shows the cross section of the model, and its total length, width and height are 1.8m, 1.08m and 0.091m, respectively. Note that, according to the original wind-tunnel test report (Song et al., 1998), Jingsha Bridge indeed met the problem of soft flutter under wind attack angle -3° – 3° , but a PC bridge deck usually does not have the problem of VIV due to its relatively high structural damping ratio. In fact, the investigated twin-side-girder section in Fig. 2 is treated as a basic deck shape among the family of two-edge girders, for which the edge girder could be box-type, I-type and circular type as summarized by Daito et al. (2002). The investigated twin-side-girder section is a typical two-edge-box girder, which is frequently used in long-span cable-supported bridges in engineering practice and the construction material could be PC, steel and steel-concrete composite. The cable-supported bridges with two-edge girders are susceptible to both VIV and soft flutter, especially for steel and steel-concrete composite decks (Zhang et al., 2019; Liao et al., 2019). Recently, the Yingwuzhou Bridge in Wuhan, which is 4-span suspension bridge (225 m + 2 × 850 m + 225 m) over Yangtze River, underwent torsional VIV at 26th April 2020, where the wind speed was 7–8 m/s and wind angle of attack around 3° . Its deck shape is a similar flat two-edge girder ($B/D = 12.7$), which is steel-concrete composite and attached with two wind fairings to improve its aerodynamic performance.

All tests were carried out at zero angle of attack in a smooth flow field with a turbulent intensity less than 0.46%. The test cases for soft flutter and torsional VIV are listed in Table 1. The different test cases of

nonlinear flutter considered the change of structural damping ratio ξ_{s0} in a wide range from 0.074%–1.49%, while torsional VIV accounted for the variation of vibration frequency f_{t0} and structural damping ratio ξ_{s0} . Because VIV often occurs at much lower wind speed than flutter, VIV tests need larger velocity scale and higher spring stiffness than flutter tests. In this connection, the tests were carried out separately for VIV and soft flutter with different sets of helical springs. Note that the structural damping ratios ξ_s were obtained via free decay responses in still air and distributed within a certain range instead of being a constant value as shown in Appendix A. This is because that the spring-suspended vibration system is actually nonlinear and structural damping ratios are thus amplitude-dependent (Gao and Zhu, 2015). After taking into account the 1/3 mass of helical springs, the total equivalent mass and mass moment of inertia are, respectively, 27.921 kg and 3.939 kg m² for flutter test F1, and 27.256 kg and 3.798 kg m² for torsional VIV test V1.

3.2. Observed VIV and soft flutter

The observed nonlinear aerodynamic instabilities, i.e., torsional VIV and soft flutter, are shown in Figs. 3–6. As shown in Fig. 3(a) and (b), the sectional model was found to undergo torsional VIVs at low range of U^* and soft flutters beyond the linear flutter boundary. Torsional VIVs were limited to two ranges of U^* , i.e., 0.07–0.13 and 0.14–0.23, while soft flutter occurred when $U^* > 0.43$. The observed amplitude of torsional VIV was characterized by a hump and that of soft flutter exhibited approximately an unlimited linear increase with U^* .

As shown in Fig. 4 and Fig. 5, the vibration responses of the sectional model were featured as a LCO for both the torsional VIV and the soft flutter. The observed LCOs always developed to stable amplitude after an initial excitation or artificial stabilization at different U^* , therefore, the observed nonlinear vibration were stable LCOs. Also note that the heaving vibration was slight when compared with the corresponding torsional one, which is consistent with previous experiments (Gao et al., 2018). The observed VIVs and soft flutters can then be simplified as a SDOF nonlinear vibration by neglecting the heave-torsion coupling effect.

Fig. 6 shows the amplitude spectrum of measured torsional angle during the stable amplitude stages of soft flutter ($U^* = 0.611$) and VIV ($U^* = 0.199$). One can find that the fundamental frequency took the most significant portion of torsional responses and the even-order harmonic

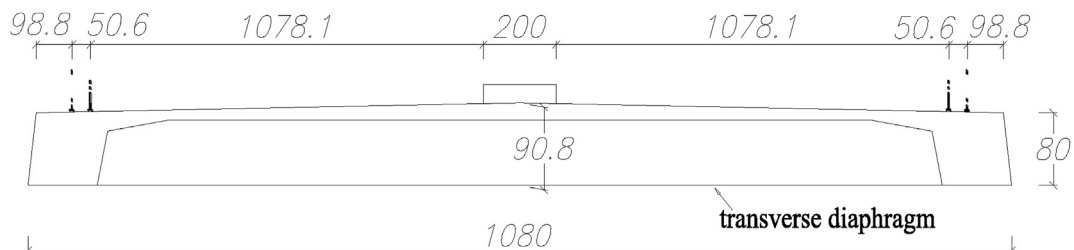


Fig. 2. Cross section of the sectional model (Unit: mm).

Table 1

Dynamic properties of test cases for soft flutter and torsional VIV. f_{t0} is the linear torsional frequency in still air, ξ_{s0} is the linear structural damping ratio of the torsional mode in still air. J_m is the mass moment of inertia of the spring-suspended vibration system.

Flutter tests				VIV tests			
#	f_{t0} (Hz)	ξ_{s0} (%)	J_m (kg·m ²)	#	f_{t0} (Hz)	ξ_{s0} (%)	J_m (kg·m ²)
F1	1.773	0.18	3.939	V1	3.168	0.096	3.798
F2	1.780	0.81	3.910	V2	2.918	0.085	4.637
F3	1.782	1.25	3.901	V3	3.162	0.32	3.807

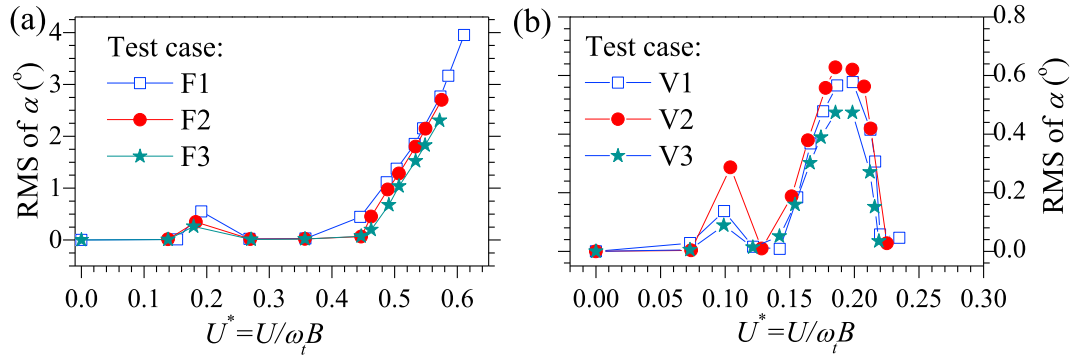


Fig. 3. The measured root mean square (RMS) values of torsional angle during (a) soft flutter and (b) VIV.

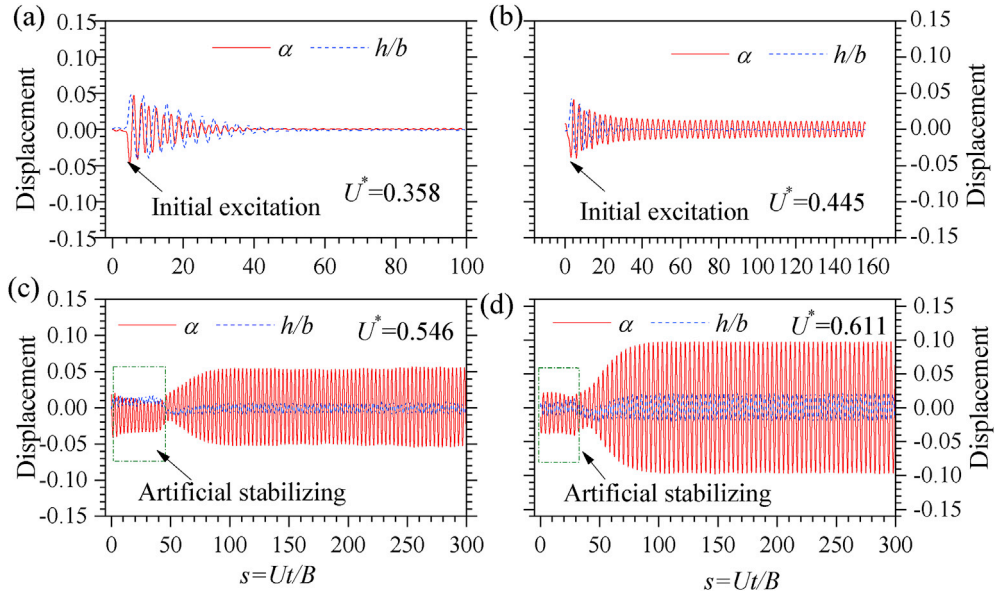


Fig. 4. Typical vibration responses during the build-up process of soft flutter. α and h represent torsional and heaving displacement, respectively. $b = B/2$ is the half width of the bridge deck.

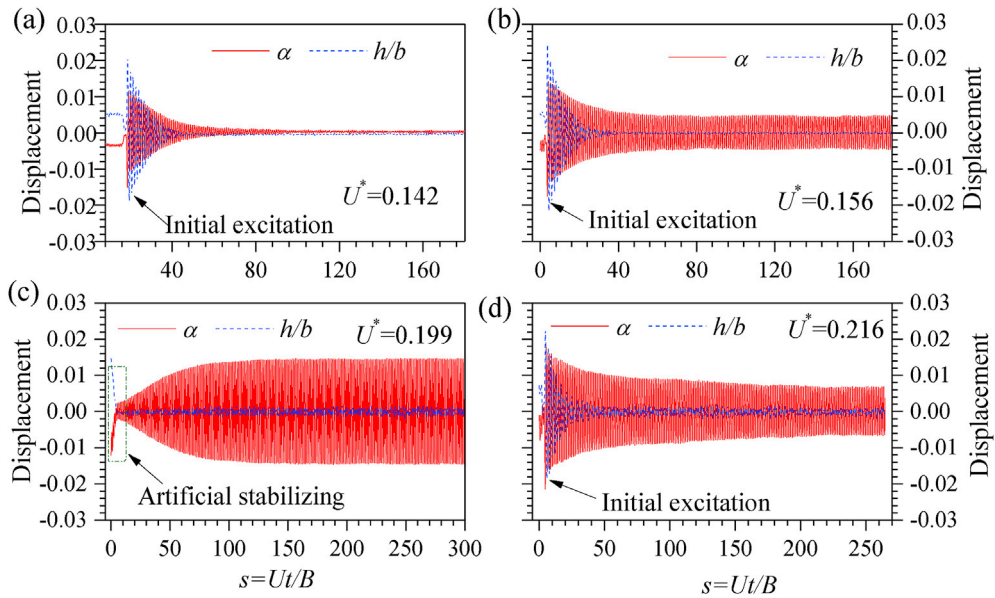


Fig. 5. Typical vibration responses during the build-up process of VIV.

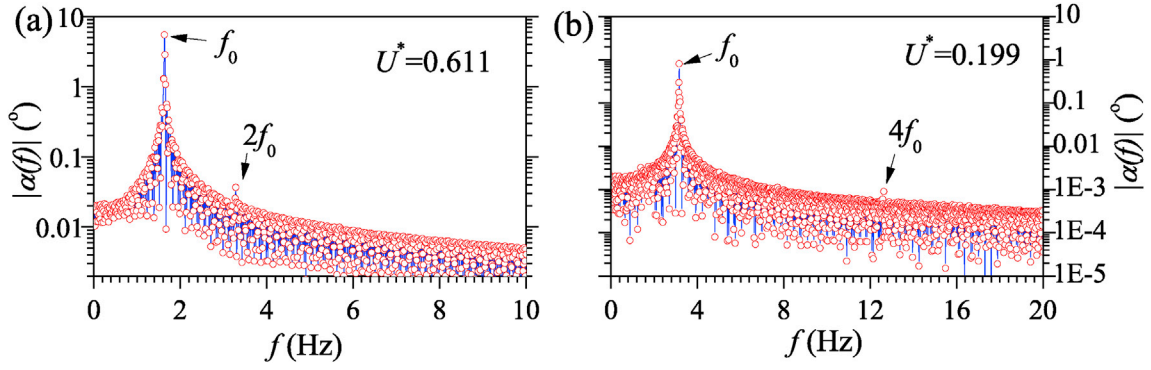


Fig. 6. Amplitude spectra of measured torsional angle during (a) soft flutter (b) VIV at a steady-amplitude stage.

components were very small, the 2nd-order and 4th-order harmonics were about 0.67% and 0.11% of fundamental parts for soft flutter and VIV, respectively. Therefore, it is appropriate to simplify the nonlinear torsional response by only keeping the fundamental-frequency component in the asymptotic solution in Eq. (9).

Finally, it is worthy to compare the vibration responses of different test cases. One can find from Fig. 3 that increasing structural damping ratio leads to obvious change of RMS values for VIV, whereas, increasing structural damping has limited effect on the RMS values of soft flutter. The difference of the RMS values during soft flutter is displayed in Fig. 7(a) by the relative stable-amplitude ratios γ , where γ is defined as the ratio of RMS values of test case F2 or F3 to that of case F1. One can find that the RMS values (or, equivalently, stable amplitude) of torsional angle decrease for case F2 and F3 with larger structural damping ratios. Also note that the influences of structural damping ratios on the RMS values diminish as reduced velocity U^* increases, which corresponds to a larger oscillation amplitude. Therefore, the vibration RMS of soft flutter is relatively insensitive to the variations of structural damping when compared with VIV, which is due to the fact that the large-amplitude oscillation during soft flutter is induced by a relatively large negative aerodynamic damping. The contribution of structural damping to the stable amplitude thus becomes smaller when it is balanced with the negative/positive aerodynamic damping in a steady-amplitude stage. As a result, the change of stable amplitude due to the variation of structural damping is relatively small and will continue to decrease as reduced velocity increases, because the negative aerodynamic damping increases with reduced velocity as indicated by the identified A_2^* in Fig. 13. Whereas, when the variations of structural damping continue to increase as large as 1%–5% which is comparable to aerodynamic damping, such a large variation of structural damping will also induce obvious change of stable amplitudes of soft flutter as suggested by Ying et al. (2017). The

above phenomenon was also confirmed in the observed unsteady galloping of rectangular cylinders by Mannini et al. (2014) and Gao et al. (2020). Because unsteady galloping is also a large-amplitude instability in the across-flow direction.

Fig. 7(b) shows the typical soft flutter responses around $U^* = 0.503$ for test cases F1~F3. It can be seen that there are obvious difference in the build-up processes for different test cases. Increasing structural damping ratio results in a wider build-up process and a smaller stable amplitude.

3.3. Measurement of nonlinear self-excited force

The technique of synchronous measurement of aerodynamic force and dynamic displacement was employed in this study to obtain the signals of aerodynamic force during VIV and soft flutter (Zhu et al., 2017; Gao et al., 2018). As shown in Fig. 8, the aerodynamic force was measured together with displacement by four three-component force balances, which were installed inside the sectional model. The force balances were piezoelectric type with a high sensitivity and small size of $0.035 \times 0.05 \times 0.05$ m. To reduce the portion of inertia force and hence improve the precision, the model was designed to be comprised of an internal rigid frame and three segments of external ‘coat’ simulating the deck shape. The middle ‘coat’ segment was separated from the rest of the sectional model and fixed onto the main rigid frame through four balances. In this way, only the dynamic force acting on the middle ‘coat’ segment was transferred to the force balances, and the inertia force contained in the measured force signals were thus significantly reduced, the proportion of aerodynamic force in the total measured dynamic force was increased obviously. Two 1–2 mm gaps were set between the middle ‘coat’ segment and two side segments to avoid any interference. The arrangement of two side segments was to reduce the unfavourable

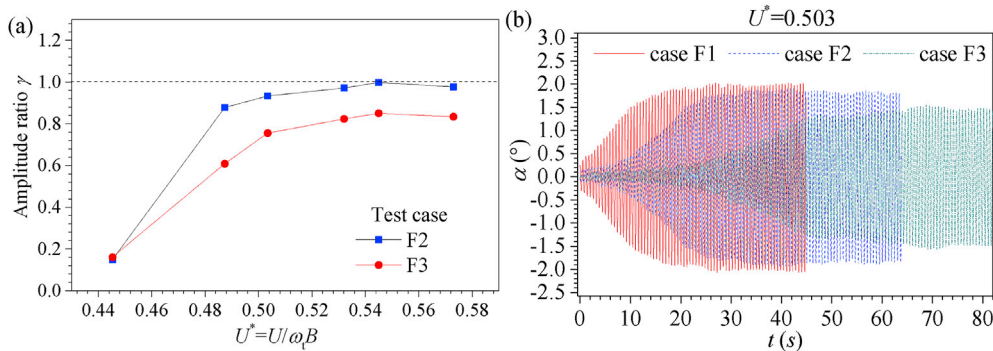


Fig. 7. Comparison of torsional response during soft flutter. (a) Stable-amplitude ratio relative to case F1; (b) torsional angle in a build-up process.

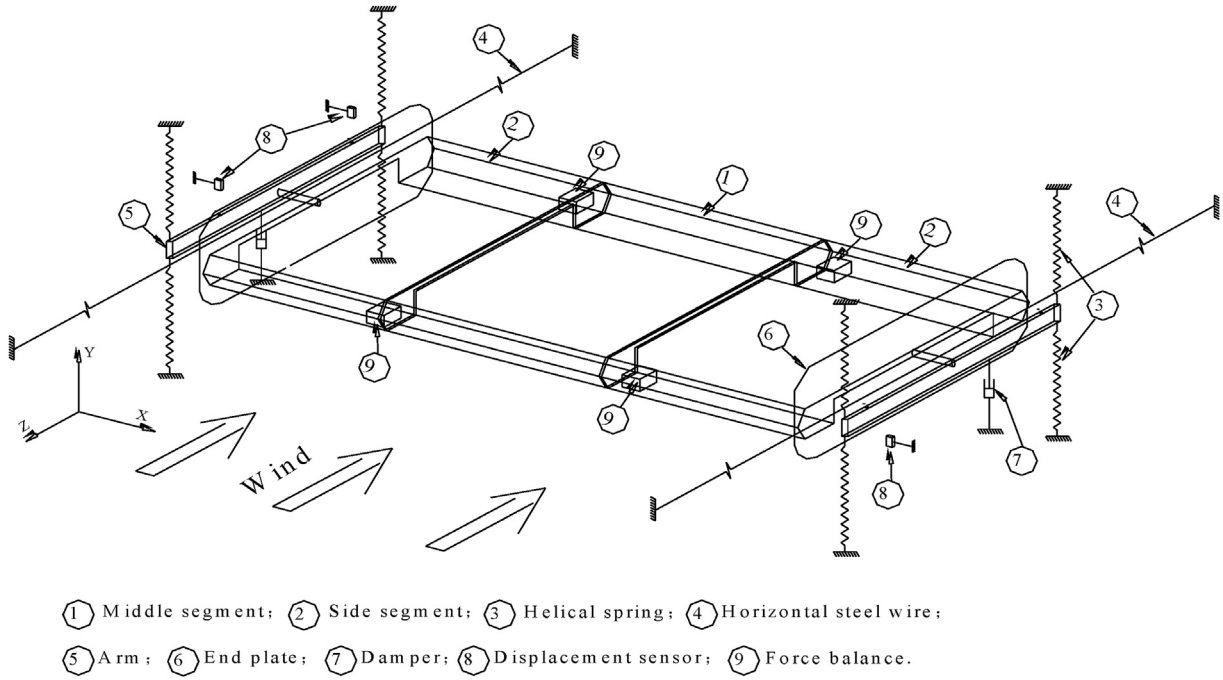


Fig. 8. General arrangement of a novel measurement technique of nonlinear aerodynamic force (Gao et al., 2018).

interference effect of boundary flows near the model ends on the force measurement. The middle 'coat' segment is 0.8m long while each of the two side coat segments is 0.5m long. The middle 'coat' was made by using light-weight wooden plates which were stiffened by thin-walled duralumin to ensure its stiffness. As a result, the mass and mass moment of inertia of the middle 'coat' are 5.1 kg and 0.282 kg m², respectively. The extraction of aerodynamic torque was briefly introduced in Appendix B.

Also note that the force sensors only measure the aerodynamic force acting on the middle segment. The aerodynamic force acting on the whole sectional model can be obtained by multiplying the measured aerodynamic force per unit length by the total spanwise length, because the flow around the sectional model can be regarded as two dimensional, the aerodynamic force is thus thought to be uniformly distributed along the length of sectional model.

4. Results and discussion

4.1. Measured nonlinear aerodynamic force

Fig. 9 and Fig. 10 illustrate the time history of the measured self-excited torque during soft flutter and VIV LCOs. The pure self-excited torque account for about 36% and 8.5% of the total measured dynamic force acting on the force balances for soft flutter at $U^* = 0.611$ and torsional VIV at $U^* = 0.187$. One can find from Fig. 9 that the curves of M_{se} were slightly distorted from sinusoidal ones, especially for that of VIV, which were due to the existence of high-order harmonics in the spectra (Fig. 10). The number of high-order harmonics was more abundant for soft flutter than for torsional VIV due to the fact that the vibration amplitude of soft flutter was larger. The 2nd and 3rd order harmonics were about 5.94% and 0.84% of the fundamental component for soft flutter at $U^* = 0.611$ and for torsional VIV at $U^* = 0.187$, were

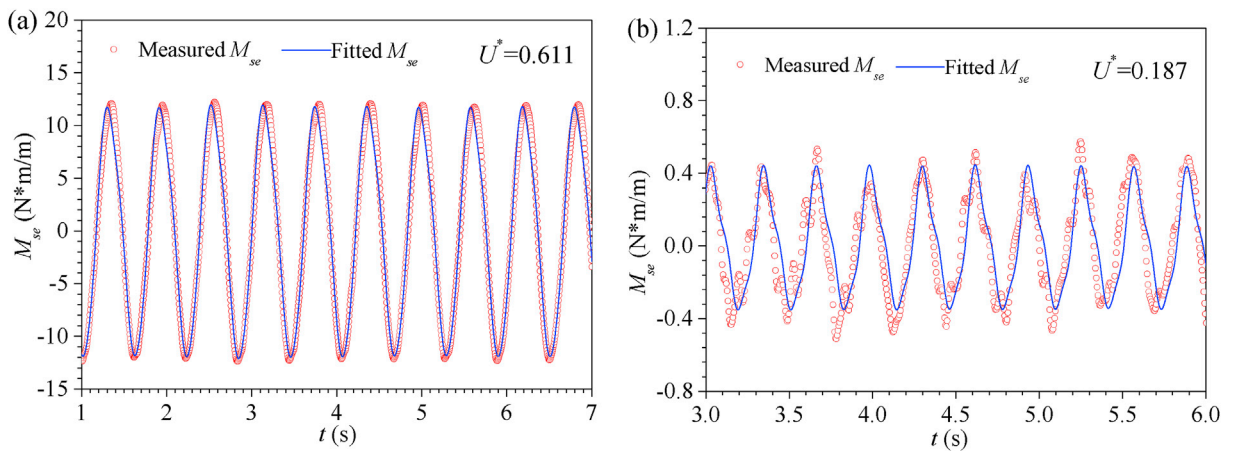


Fig. 9. Comparison of the measured and fitted time history of M_{se} by Eq. (20) during (a) soft flutter and (b) VIV.

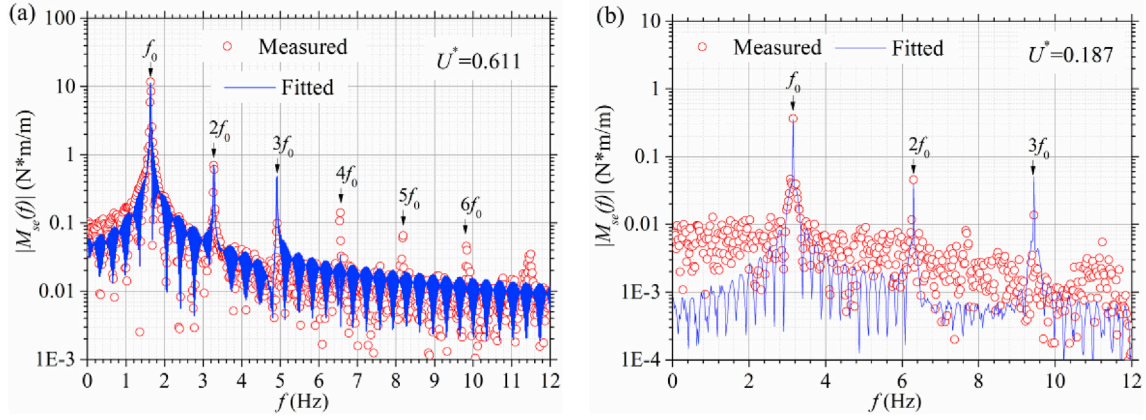


Fig. 10. Amplitude spectra of the measured M_{se} and the fitted results by Eq. (20) during (a) soft flutter and (b) VIV at steady-amplitude stage.

about 12.5% and 3.8% indicating that aerodynamic nonlinearity was stronger for torsional VIV.

4.2. Validation of measured aerodynamic force

The accuracy of the above measured $M_{se}(t)$ needs to be validated prior to be substituted into Eq. (13)~ Eq. (14) to identify aerodynamic parameters. The validation of measured M_{se} was carried out by solving the non-autonomous governing equation of motion as follows

$$I\ddot{\alpha} + \tilde{c}_m(a_\alpha)\dot{\alpha} + \tilde{k}_m(a_\alpha)\alpha = \tilde{M}_{se}(t) \quad (18)$$

where I is the equivalent mass moment of inertia per unit length of the oscillatory system; $\tilde{c}_m(a_\alpha) = 2I\omega_0\tilde{\xi}_s(a_\alpha)$ and $\tilde{k}_m(a_\alpha) = 4\pi^2 I f_t^2(a_\alpha)$ are the amplitude-dependent structural damping and stiffness coefficients in the torsional mode, which were identified from the free-decay responses in still air and displayed in Appendix A. $\tilde{M}_{se}(t)$ represents the time history of measured self-excited torque per unit length. a_α is the transient torsional amplitude and can be determined by:

$$a_\alpha(t) = \sqrt{\alpha(t)^2 + (\dot{\alpha}(t)/\omega_{t0})^2} \quad (19)$$

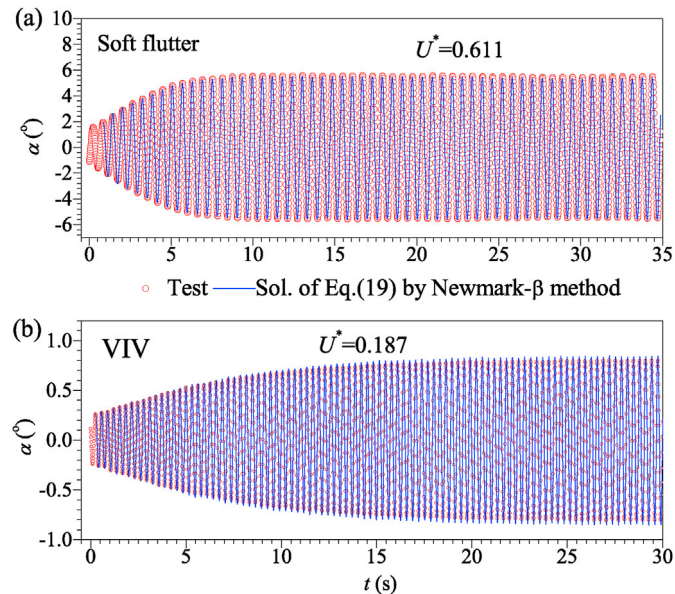


Fig. 11. Comparison between the measured and calculated torsional angle during soft flutter (case F1) and VIV (case V1).

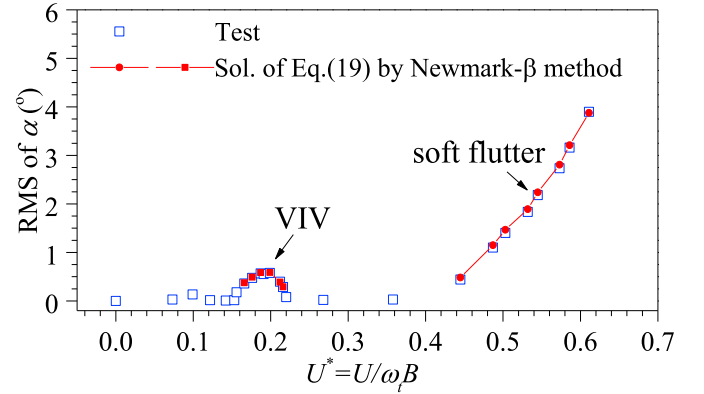


Fig. 12. Comparison between the measured and calculated RMS values of torsional angle, where soft flutter and VIV are corresponding to test case F1 and V1, respectively.

where $\omega_{t0} = 2\pi f_{t0}$ is the linear torsional circular frequency in still air.

The solving of Eq. (18) was carried out by Newmark- β method in conjunction with calculating the transient amplitude by Eq. (19). The calculated results are shown in Fig. 11 and Fig. 12. One can find that the calculated time histories and RMS values of torsional angle both agreed very well with experiments for soft flutter and torsional VIV. Hence, the measured M_{se} was verified to have a good accuracy and reliability.

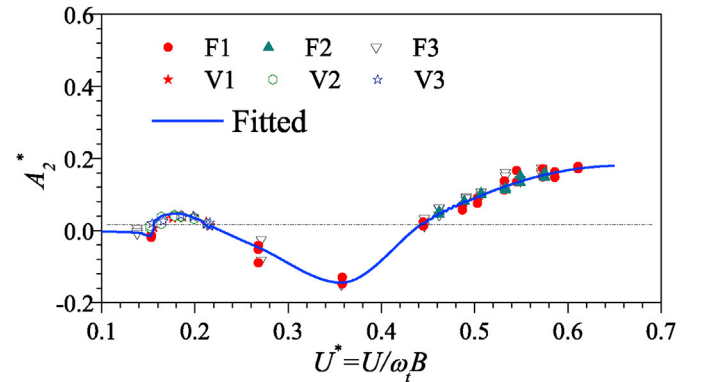


Fig. 13. Identified aerodynamic linear damping parameters A_2^* from different test cases.

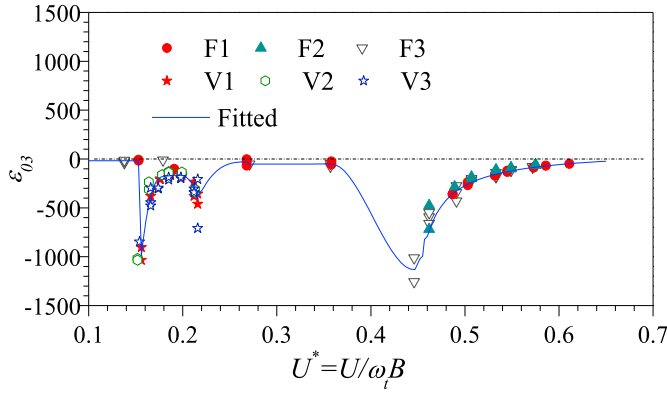


Fig. 14. Identified aerodynamic nonlinear damping parameters ε_{03} from different test cases.

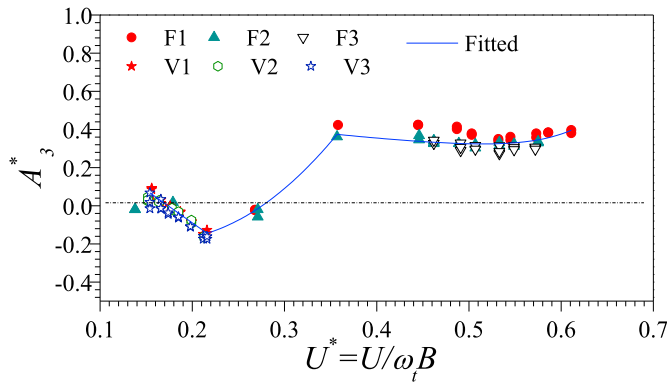


Fig. 15. Identified aerodynamic stiffness parameters A_3^* from different test cases.

4.3. Identified aerodynamic parameters

The aerodynamic parameters in the proposed model as Eq. (6) were identified via the energy-balanced-principle in Section 2.3 by substituting the measured $M_{se}(t)$ and $\alpha(t)$ into Eq. (13) and Eq. (14). The identified aerodynamic parameters A_2^* , ε_{03} and A_3^* are displayed in Figs. 13–15. One can find that the identified parameters from different test cases clustered around a clear trend and the scattering were limited in spite of the variations of structural properties of different test cases in Table 1. Also note from Figs. 3 and 7 that the vibration amplitudes of different test cases were different, therefore, the identified aerodynamic parameters were also insensitive to the variations of motion amplitude. Therefore, the aerodynamic nonlinearity has been adequately considered by the proposed model and the corresponding aerodynamic parameters are only dependent on the aerodynamic shape of the tested section and reduced velocity. From an engineering point of view, it is a precondition for the aerodynamic parameters to be insensitive of motion amplitudes, structural properties, when apply the wind-tunnel test results to prototype bridges.

The energy evolution mechanisms of soft flutter and torsional VIV can be inferred from the variation behaviors of parameters A_2^* and ε_{03} with reduced velocity U^* . As shown in Fig. 13, the parameter A_2^* is positive during VIV and soft flutter, which indicates that the corresponding linear aerodynamic damping torque in Eq. (6) does positive work on the vibration system and thus provides a negative damping effect. The A_2^* monotonously increases with U^* after the onset point of linear flutter,

which results in an unlimited increase of the soft flutter steady amplitude with U^* . When the U^* is outside the range of VIV and soft flutter, A_2^* is negative indicating that the linear term in Eq. (6) does negative work and thus provides a positive aerodynamic damping effect.

The variation of ε_{03} with reduced velocity U^* in Fig. 14 behaves in a different pattern from that of A_2^* . As can be seen, ε_{03} is always negative in total tested range of U^* , which means that the corresponding nonlinear aerodynamic damping terms $\dot{\alpha}^3$ in Eq. (6) always does negative work on the vibration system during each vibration period and hence provides a nonlinear positive damping effect. Also note that ε_{03} exhibits a sharply decrease phenomenon around the onset point of the torsional VIV and the soft flutter, and gradually increase to zero thereafter. The inherent flow mechanism of this phenomenon is not clear yet and needs further investigation, but it is probably due to that the fluctuating aerodynamic force is significantly enhanced in the synchronized region as suggested by Wilkinson (1981a, 1981b).

In summary, the linear aerodynamic damping component $\dot{\alpha}$ provides a negative damping effect and pumps energy into the vibration system during both the VIV and the soft flutter. Meanwhile, the cubic nonlinear aerodynamic term $\dot{\alpha}^3$ provides a significant positive aerodynamic damping and thus is the inherent factor leading to self-limiting phenomena, which is also consistent with the case of side-ratio $B/D = 7.8$ in the previous study (Gao et al., 2018). Therefore, torsional VIV and soft flutter share the same nonlinear damping mechanism during the development of LCO, although the micro-mechanism of flow field may be different.

Also note that the time history of measured $M_{se}(t)$ cannot be adequately fitted by the proposed model as Eq. (6), because the empirical model only contains three terms $\dot{\alpha}$, $\dot{\alpha}^3$ and α to consider the aerodynamic damping and stiffness effect of $M_{se}(t)$. Other motion-induced ‘pure force’ terms were needed in Eq. (6) to better fit the measured $M_{se}(t)$. After introducing ‘pure force’ terms into Eq. (6), the full-order nonlinear model is

$$M_{se}(\alpha, \dot{\alpha}) = \rho U^2 B^2 \left[K A_2^* \left(1 + \varepsilon_{03} \left(\frac{B \dot{\alpha}}{U} \right)^2 + \varepsilon_{02} \frac{B \dot{\alpha}}{U} + \varepsilon_{11} \alpha \right) \frac{B \dot{\alpha}}{U} + K^2 A_3^* (1 + \varepsilon_{20} \alpha) \alpha + \frac{1}{2} \tilde{C}_M \sin(K_{vs} s + \varphi) \right] \quad (20)$$

where ε_{02} , ε_{11} and ε_{20} are aerodynamic parameters and similar to ε_{03} , are functions of reduced velocity U^* . The fitted results of $M_{se}(t)$ by Eq. (20) are shown in Figs. 9 and 10, one can find that both the time history and amplitude spectra can be fitted rather well. Note that the 2nd-order harmonics in Fig. 10 was fitted by the newly-added quadratic terms in Eq. (20).

As demonstrated in the previous study (Gao et al., 2018), the motion-induced ‘pure force’ terms do not contribute to aerodynamic work or reactive work, therefore the full-order model as Eq. (20) are equivalent to Eq. (6) in the sense of aerodynamic damping and stiffness effects. Since the steady amplitude of torsional VIV and soft flutter are the main concern in engineering practice, the ‘pure force’ terms in Eq. (20) can be thus dropped and Eq. (20) is simplified as Eq. (6) without affecting the predicted aeroelastic responses. It is also worthy to point out that the motion-induced ‘pure-force’ terms vary with side ratio, because for the case of $B/D = 7.8$ in the previous study, the motion-induced ‘pure-force’ terms were $\dot{\alpha}^2$ and α^2 which are different from the terms $\dot{\alpha}^2$ and $\alpha \dot{\alpha}$ in Eq. (20).

4.4. Numerical validation and discussion

The variations of aerodynamic parameters with reduced wind speed U^* in the proposed model were obtained by fitting the identified results

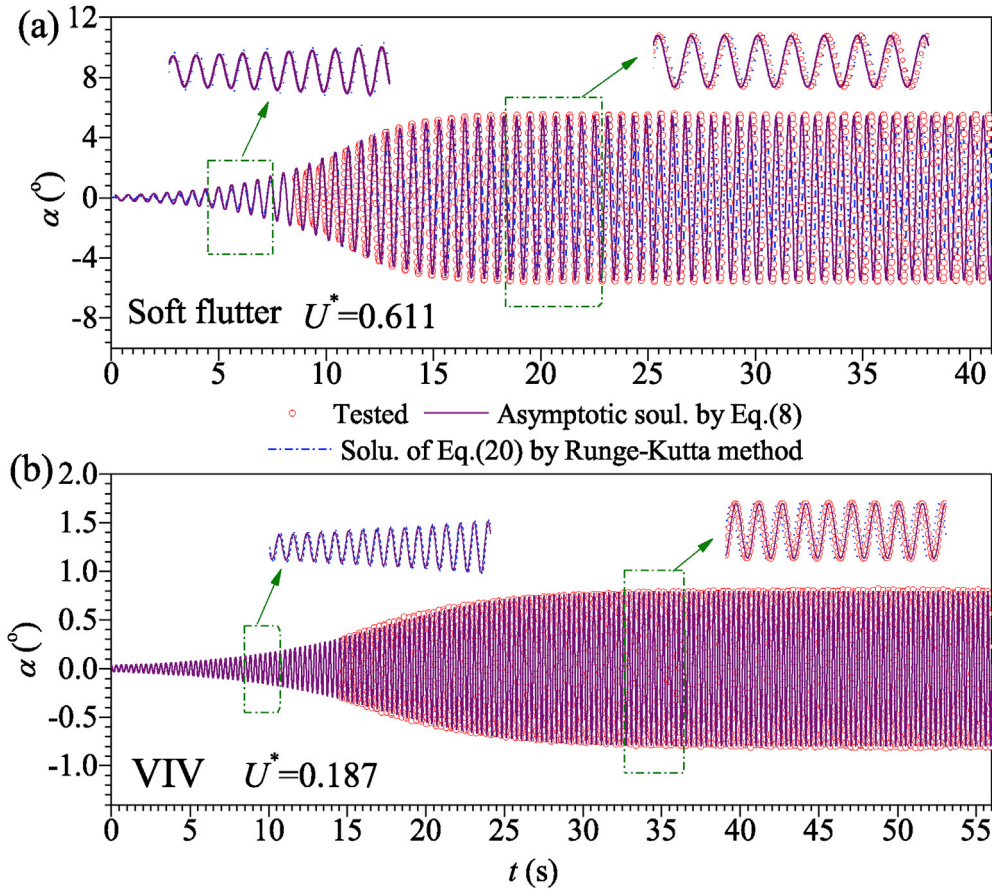


Fig. 16. Comparison between the measured and calculated torsional angle during (a) soft flutter and (b) VIV. The responses of VIV and soft flutter are corresponding to test case V1 and F1, respectively.

by smooth curves as shown in Figs. 13–15. The feasibility of the proposed empirical model can then be validated by calculating the VIV and soft flutter responses by the fitted aerodynamic parameters and compared with experimental results. The governing equation of motion for self-excited vibration under soft flutter and torsional VIV is then written as follows:

$$I\ddot{\alpha} + \tilde{c}_m(a_\alpha)\dot{\alpha} + \tilde{k}_m(a_\alpha)\alpha = \rho U^2 B^2 [K\tilde{A}_2^*(1 + \tilde{\varepsilon}_{03}\dot{\alpha}^2)\dot{\alpha} + K^2\tilde{A}_3^*\alpha] \quad (21)$$

where $\tilde{A}_2^*$, $\tilde{\varepsilon}_{03}$ and $\tilde{A}_3^*$ are the fitted aerodynamic parameters by smooth curves in Figs. 13–15.

The solving of Eq. (21) was performed through a step-by-step iteration procedure using the standard 4th-order Runge-Kutta method in conjunction with calculating the transient amplitude by Eq. (19) to update amplitude-dependent parameters in Eq. (21).

Fig. 16 shows the comparisons between the measured and predicted responses of soft flutter at $U^* = 0.611$ and torsional VIV at $U^* = 0.187$, respectively. It can be found that the proposed empirical model can predict the transient amplitudes of both the torsional VIV and the soft flutter responses rather well. However, some discrepancies between the transient phases of the calculated and measured responses can be observed, which is due to the neglecting of nonlinear aerodynamic stiffness effect of M_{se} in the proposed empirical model. Since the achievable steady amplitudes of VIV and soft flutter instabilities are the primary concern in engineering application, only the nonlinearity in aerodynamic damping was modeled and the nonlinear aerodynamic stiffness was neglected in the proposed model.

Furthermore, the comparisons between the measured and calculated torsional RMS values are shown in Fig. 17 for the test cases F1 and V1. One can find that the calculated torsional RMS agrees well with experimental one.

Finally, one can find from Figs. 16 and 17 that the predicted responses by the asymptotic solution derived in Section 2.2 were in good agreement

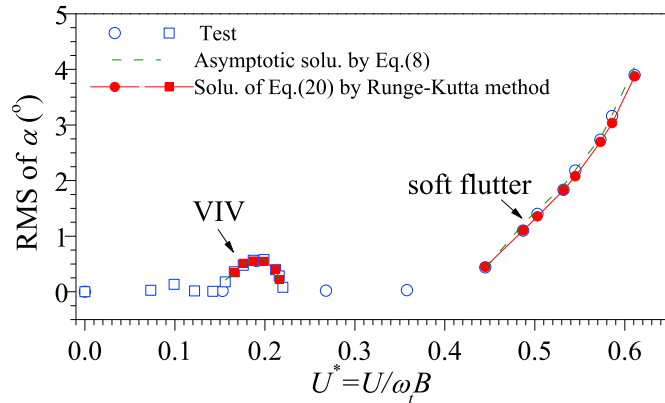


Fig. 17. Comparison between the measured and calculated RMS values of torsional angle. The VIV and soft flutter are corresponding to test case V1 and F1, respectively.

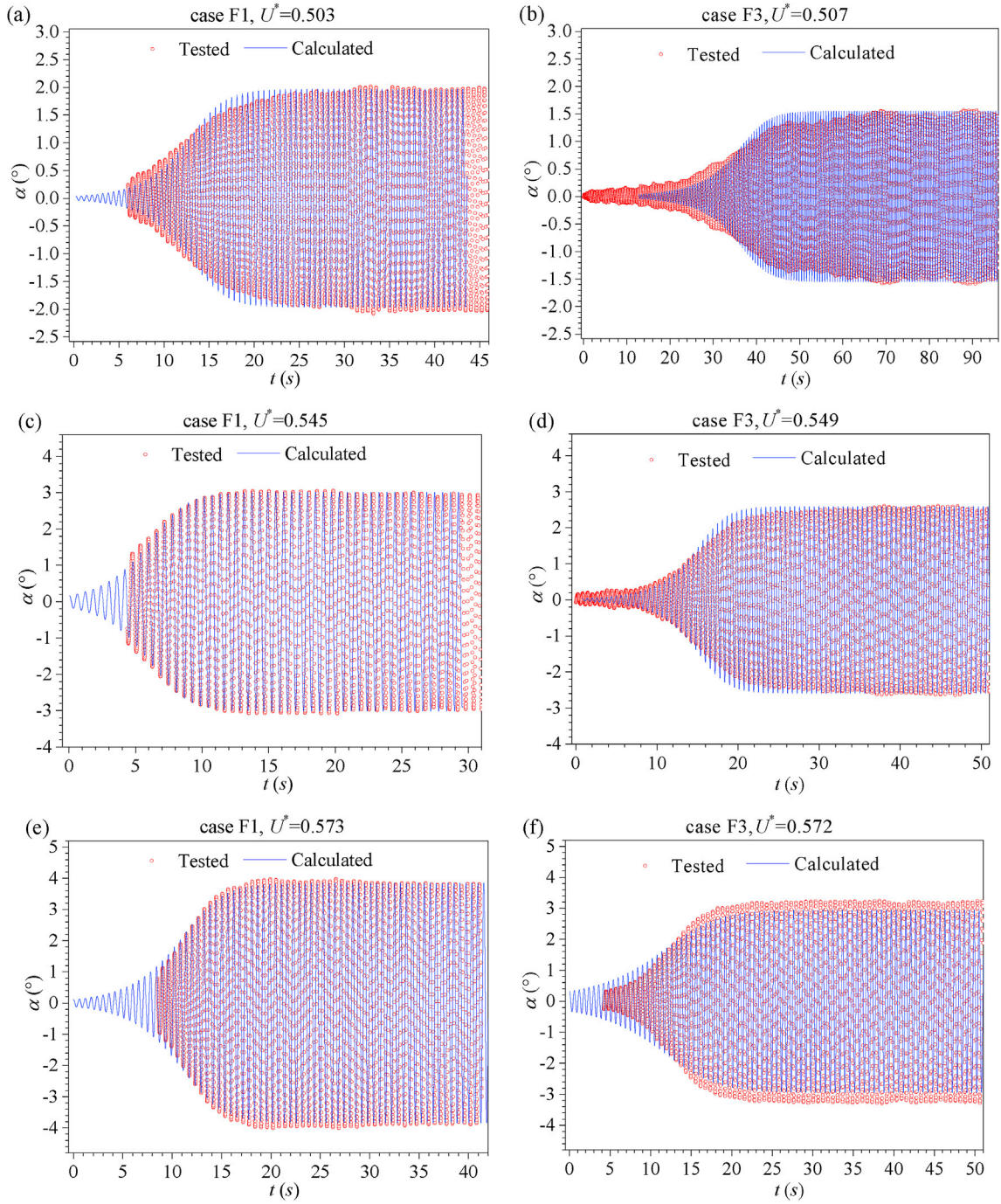


Fig. 18. Prediction of torsional angle during soft flutter for case F1 and F3.

with the numerical solution of Eq. (21) through 4th-order Runge-Kutta method, therefore, the asymptotic solution in Eq. (9) can be directly used in a quick estimation of torsional vibration responses of both soft flutter and torsional VIV.

Fig. 18 shows the comparison of the calculated soft flutter LCOs with the tested ones for case F1 and F3. One can find that the proposed empirical model and the identified parameters in Figs. 13–15 can satisfactorily predict the build-up process and stable-amplitudes of soft flutter LCOs under different U^* for test case F1 and F3, in spite that their vibration responses are clearly different. The discrepancy becomes larger

for case F3, which mainly lies in the build-up process and that in stable amplitude is relatively small. From an engineering point, the stable amplitude is the main concern and its largest discrepancy is about -9.5% at $U^* = 0.572$. Therefore, the proposed model and its identified parameters can be applied to different structural properties.

Similar to Figs. 16–18, the vibration responses of torsional VIV and soft flutter under other tests cases in Table 1 can also be satisfactorily predicted by adopting the corresponding structural parameters on the left side of Eq. (21). Therefore, one can find that the proposed empirical model can be applied to different structural conditions. The proposed

empirical model has thus been preliminarily validated. Undoubtedly, the discrepancy of case F3 in Fig. 18 still needs to be reduced. It is a future work of this study to validate the proposed empirical model in a wider structural properties and improve the estimation accuracy of aerodynamic parameters to reduce the predicting precision.

4.5. Discussion on the applicability to vertical VIV

The proposed empirical model as Eq. (6) was employed to both SDOF soft flutter and torsional VIV through above-mentioned wind-tunnel tests. The proposed empirical model can be also extended to the vertical degree of freedom as

$$L_{se}(h, \dot{h}) = \rho U^2 D \left[Y_1^* \left(1 + \varepsilon_{03} \frac{\dot{h}^2}{U^2} \right) \frac{\dot{h}}{U} + Y_3^* \frac{h}{D} + \frac{1}{2} \bar{C}_L \sin(K_{vs}s + \varphi) \right] \quad (22)$$

where Y_1^* and Y_3^* are K -dependent aerodynamic parameters. D is the depth of the investigated bridge section.

Zhu et al. (2017) recently pointed out through wind-tunnel tests that the motion-induced force in Eq. (22) can be used as a universal nonlinear vortex-induced-force model for predicting the stable amplitudes of vertical VIV with satisfactory accuracy for three different bridge decks, which are a fully closed box deck, a semi-closed box deck and a centrally slotted box deck. Therefore, the proposed empirical model as Eq. (6) shares a similar nonlinear form with the universal vortex-induced force of vertical VIVs.

5. Conclusions

Although flutter and VIV are conventionally treated as two different aerodynamic instabilities of flexible bridges, they in fact share many features in common. The possibility of building a universal empirical model of nonlinear self-excited force was investigated in this study. In the previous study, a new empirical model was proposed based on the observed soft flutter of a twin-side-girder section with a medium side-ratio as 7.8; the empirical model was firstly refined in this study by considering the pure vortex shedding force and extended to both torsional VIV and soft flutter for a general bluff section with a large frontal area, such as a general twin-side-girder bridge deck or a π -shape section. Based on the empirical model, a first-order asymptotic nonlinear solution of the universal self-excited vibration was deduced and an improved parameter identification method was then developed based on an “energy-balanced” principle. A series of elastically-mounted sectional model tests on a flat twin-side-girder bridge deck were carried out to check the feasibility of the proposed model in case of a larger side ratio as 11.9, which is more common in long-span bridges. It was found that the

sectional model underwent torsional VIV in low reduced wind range and soft flutter beyond linear flutter boundary; both VIV and soft flutter exhibited a self-limiting phenomenon in the time domain and a quasi-harmonic response corresponding to the fundamental frequency in the frequency domain.

The identified aerodynamic parameters from different structural damping conditions were consistent and in a clear trend for both torsional VIV and soft flutter. It was found that the nonlinear damping mechanism during the build-up of vibration amplitude was same for torsional VIV and soft flutter, and also consistent for different side ratios when compared with the previous study, whereas, the motion-induced ‘pure force’ varies with side ratio. The proposed empirical model was found to be applicable to the estimation of stable amplitudes of torsional VIV and soft flutter under different structural conditions. The proposed empirical model can be extended to vertical VIV as a universal form of vertical vortex-induced force model for three different bridge deck shapes.

The future work of this study is to check the applicability of the proposed model in a wider range of structural properties and consider different side ratios.

CRediT authorship contribution statement

Guangzhong Gao: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing - original draft, Writing - review & editing, Visualization, Funding acquisition. **Ledong Zhu:** Conceptualization, Methodology, Supervision, Project administration, Funding acquisition, Resources. **Jiawu Li:** Supervision, Project administration, Funding acquisition, Resources. **Wanshui Han:** Supervision, Project administration, Funding acquisition, Resources.

Declaration of competing interest

The authors declare that they have no conflicts of interest with respect to the study described in this manuscript.

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Appendix A. Nonlinear structural parameters of the elastically-supported system

Due to the mechanical nonlinearities in the elastically-supported oscillatory system, the structural damping ratio and frequency both exhibit amplitude-dependent behaviors under large amplitude excitation. The amplitude-dependent parameters were identified by a time-domain algorithm from the free decay responses of the sectional model in still air (Gao and Zhu, 2015). The identified results are shown in Figs. 19 and 20. Note that the different data symbols in Figs. 19 and 20 represent the identified results from different batches of initial perturbation and it can be found that the identified results showed good repeatability. The identified amplitude-dependent parameters were fitted by smooth curves and used in the solving of the governing equations as Eq. (18) and Eq. (21).

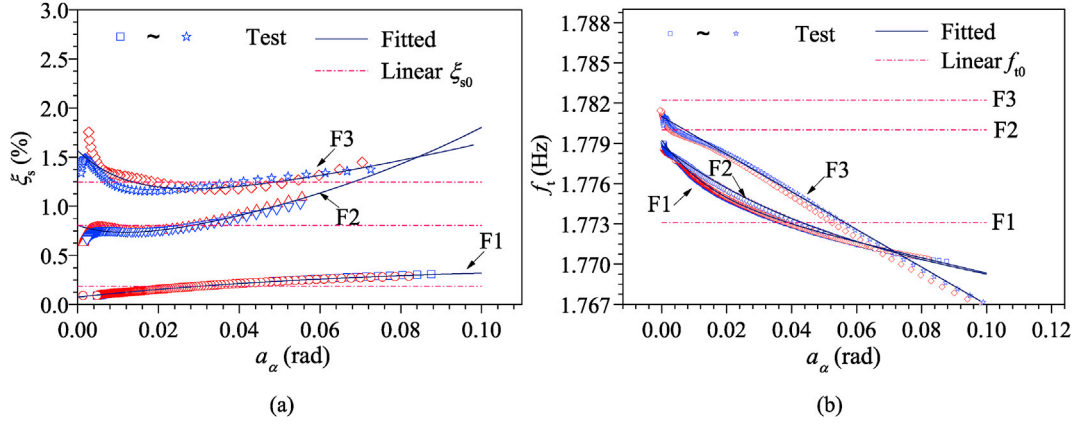


Fig. 19. Amplitude-dependent structural (a) damping ratio and (b) frequency of the spring-suspended system for test case F1~F3.

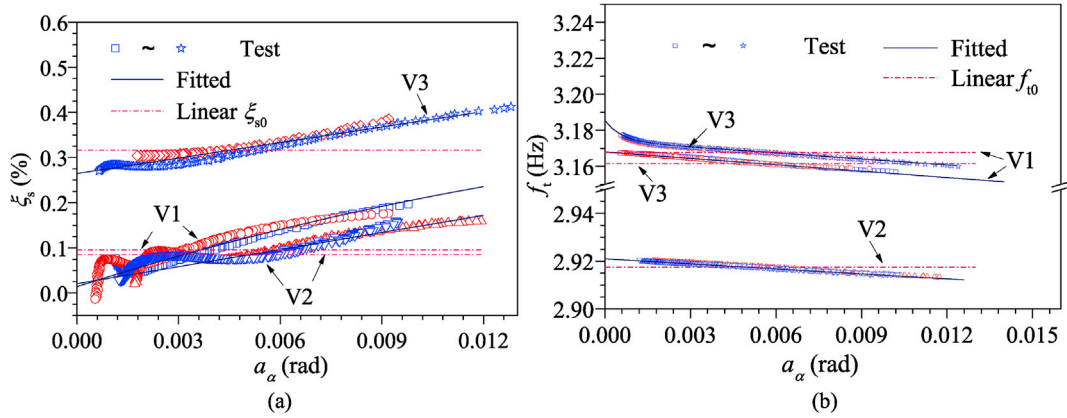


Fig. 20. Amplitude-dependent structural (a) damping ratio and (b) frequency of the spring-suspended system for test case V1~V3.

Appendix B. The extraction of aerodynamic force via the measurement technique

The loading state of the exterior ‘coat’ of the middle segment during torsional vibration is shown in Fig. 21, where the dynamic force components measured by each force sensor are demonstrated in the positive directions. According to D’Alambert’s Principle, the dynamic equilibrium equation of the exterior ‘coat’ per unit length can be expressed as

$$M_{ms}(t) = M_{se}(t) + M_{sc}^0(t) + M_I(t) + M_{vs}(t) \quad (23)$$

where $M_I = -J_s \cdot \ddot{a}$ is the inertial moment of the exterior ‘coat’ per unit length, being J_s the mass moment of inertia per unit length and $\ddot{a}$ the angular acceleration; M_{se} represents the motion-induced aeroelastic torque and M_{vs} is the pure vortex shedding force. M_{ms} is the measured dynamic torque by the 4 force sensors, which is expressed as

$$M_{ms}(t) = [(M_{m1} + M_{m2} - M_{m3} - M_{m4}) + (F_{my1} - F_{my2} + F_{my3} - F_{my4}) \times b_m / 2] / l_m \quad (24)$$

where M_{mi} , F_{myi} ($i = 1, 2, 3, 4$) are the measured dynamic torque and shear force by each force sensor, respectively; l_m is the span-wise length of the middle segment, b_m is the chord-wise distance of two force sensors.

The term $M_{sc}^0(t)$ in Eq. (23) is the non-wind-induced force (Gao et al., 2018; Chen et al., 2020), which is expressed as

$$M_{sc}^0(t) = -J_0 \cdot \ddot{a}(t) - c_0(a_\alpha) \cdot \dot{a}(t) \quad (25)$$

where J_0 represent the added moment of inertia; $c_0(a_\alpha)$ is the nonlinear added damping coefficient, which considers the nonlinear aerodynamic effect by expressing as a function of transient torsional amplitude a_α . Both J_0 and c_0 were identified from the measured force signals during the free decay process in still air.

The aerodynamic torque was extracted based on Eq. (23)~ Eq. (25), which is

$$M_{se}(t) + M_{vs}(t) = M_{ms}(t) + (J_0 + J_s) \cdot \ddot{\alpha}(t) + c_0(a_\alpha) \cdot \dot{\alpha}(t) \quad (26)$$

Also note that the pure vortex shedding force is negligible for large-amplitude vibration during VIV and soft flutter, Eq. (26) can be simplified as

$$M_{se}(t) = M_{ms}(t) + (J_0 + J_s) \cdot \ddot{\alpha}(t) + c_0(a_\alpha) \cdot \dot{\alpha}(t) \quad (27)$$

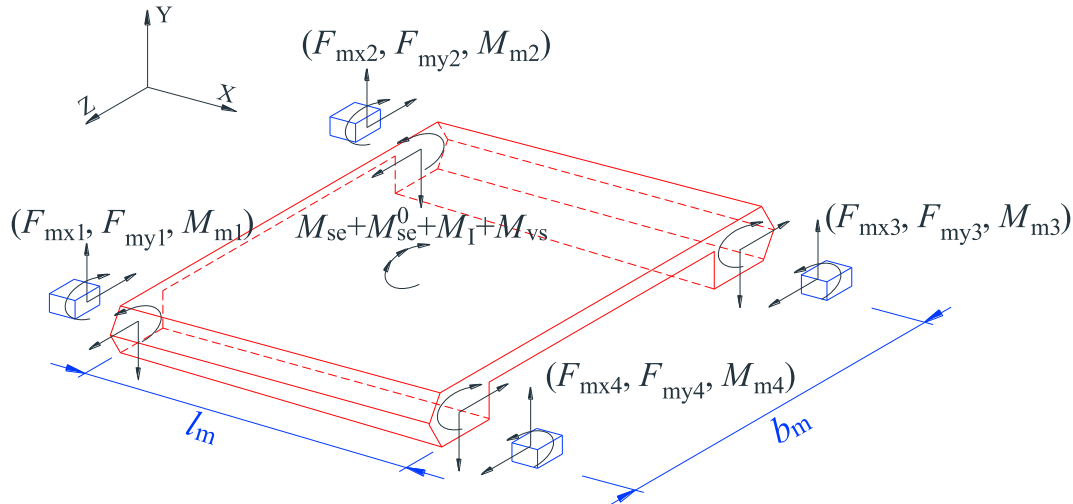


Fig. 21. Loading state of the exterior 'coat' of the middle segment during torsional vibration (Gao et al., 2018).

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